Dawood Rehman

Ph.D. Scholar, Biomedical Artificial Intelligence
Behavioral Informatics and Interaction-computation (BIIC) Lab, GEN III
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Research Interests

Computer Vision, LLMs, Generative AI, Multi-agent AI Systems, and Trustworthy AI

Education

Ph.D. in Biomedical Artificial Intelligence (BAI) 2024 – Present
National Tsing Hua University, Taiwan. Research Focus: Generative AI; Uncertainty and Trustworthy AI, Test-Time Adaptation, Neuroimaging, Brain Age Analysis.

Master of Philosophy (M.Phil) in Statistics March 2022 – April 2024
University of Malakand, Pakistan.
CGPA: 3.94/4.0.
Thesis: *Missing Values Imputation in Time Series Analysis: A Comparison of Widely Applied Data Imputation Methods.*
Supervisor: Prof. Dr. Abdul Salam.

Bachelor of Science in Statistics (BS) Sep 2017 – August 2021
University of Malakand, Pakistan.
CGPA: 3.84/4.0 (Gold Medalist).
Thesis: *Statistical Investigation of Crime Trend in District Charsadda, KP, Pakistan.*

Associate Degree in Education (ADE) Sep 2018 – August 2020
Allama Iqbal Open University, Pakistan.

Research Experience

Full Time Ph.D. Researcher Feb 2025 – Ongoing
Behavioural Informatics and Interaction Computation (BIIC) Lab
Ph.D. Program in Biomedical Artificial Intelligence, Department of Electrical Engineering, National Tsing Hua University, Hsinchu, Taiwan.

Lab Rotation (Visiting Researcher) Sep 2024 – Feb 2025
Human-Centered Machine Learning Lab (HMI)
Department of Computer Science, National Tsing Hua University, Taiwan

- Developing generative AI models for medical imaging applications.
- Working on diffusion-based models for disease detection and bias mitigation.

- Applying Bayesian deep learning for uncertainty estimation in clinical AI systems.
- Designing multimodal and transformer-based architectures for medical image analysis.

Publications

1. Dawood Rehman, Chi-Chun Lee
   DUCTA: Dynamic Uncertainty-Guided Test-Time Adaptation for Medical Image Diagnosis. ***Code***
   (MICCAI 2026, **Submitted**).
2. Dawood Rehman, Natalia Vilor-Tejedor, Chi-Chun Lee
   CMUPF: Cross-Modal Uncertainty Propagation Framework for Robust Alzheimer's Disease Classification. ***Code***
   (MICCAI 2026, **Submitted**)
3. Dawood Rehman, Huan-Yu Chen, Chi-Chun Lee
   CDC-Fair: Counterfactual Denoising Consistency for Multi-Demographic Fairness in Chest X-ray Classification. ***Code***
   (MICCAI 2026, **Submitted**)
4. Dawood Rehman, Huan-Yu Chen, Chi-Chun Lee
   Uncertainty-Conditioned Classification and Medical Report Generation. A Multi-Agent Bayesian Framework for Interpretable Chest X-ray Diagnosis. ***Code***
   (IEEE Journal of Biomedical and Health Informatics (JBHI) 2026, **Submitted**).
5. Dawood Rehman, Chi-Chun Lee
   UAM-CXR: Uncertainty-Aware Multimodal Learning for Chest X-ray Classification via Cross-Modal Conformal Prediction. ***Code***
   (IEEE EMBC 2026, **Accepted**).
6. Dawood Rehman, Huan-Yu Chen, Chi-Chun Lee
   Memory-Augmented Hierarchical Graph Networks for Interpretable Flow Cytometry Classification. ***Code***
   (IEEE EMBC 2026, **Accepted**).
7. Dawood Rehman, Huan-Yu Chen, Chi-Chun Lee, Natalia Vilor-Tejedor, Shreya Upadhyay
   Diffusion-Synthesized Chest X-rays Improve Fairness and Diagnostic Performance. ***Code***
   (PLOS Digital Health, 2026, **DOI**).
8. Dawood Rehman, Huan-Yu Chen, Chi-Chun Lee
   Bayesian Modelling for Enhanced Uncertainty, Consistency, and Calibration in Multi-Disease Chest X-Ray Diagnosis. ***Code***
   (IEEE ICASSP, 2026, **Accepted**).
9. Ullah, U., Rehman, D., Khan, S., Rashid, H., & Ullah, I. (2024).
   Forecasting Foreign Exchange Rate with Machine Learning Techniques: Chinese Yuan to US Dollar Using XGBoost and LSTM Models.
   (iRASD Journal of Economics, 2024)

Awards and Honors

- Young Scholars Research Grant, Research Development and Innovation Authority, Kingdom of Saudi Arabia (2024).
- Chiang Mai University (CMU) Presidential Scholarship, Thailand (2023).
- Gold Medal, B.S. in Statistics, University of Malakand (2022).
- Ehsaas Undergraduate Merit-Based Scholarship.
- Graduate Academic Excellence Award (CGPA 3.94/4.0).
- IELTS Overall Band Score: 6.5 (Speaking: 7.5).

Teaching and Professional Experience

Statistical Data Analyst and Teaching Assistant 2023 – 2024
University of Malakand, Pakistan

- Project: Statistical Investigation of Crime Trend in District Charsadda, KP, Pakistan.
- Assisted in teaching statistics courses and supervised student projects.
- Data Analytics and Business Intelligence, DigiSkills Pakistan (2022).

Freelance Data Scientist
Worked on statistical modeling, forecasting, and machine learning projects.

Technical Skills

Machine Learning & AI: Diffusion Models, Deep Learning, Bayesian Methods, Uncertainty Estimation, Multimodal Learning, Transformer Architectures

Statistical Methods: MCMC Simulation, Time Series Analysis, Forecasting, Prediction, Missing Data Imputation

Programming: C, Python (PyTorch, TensorFlow), R

Other: Data Integration, Model Fine-Tuning, Model Training and Deployment, Image Processing, Disease Detection and Classification

References

Professor Chi Chun Lee
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Department of Electrical Engineering, Ph.D. Program in Biomedical Artificial Intelligence, National Tsing Hua University, Hsinchu, Taiwan
Email: cclee@ee.nthu.edu.tw

Professor Natalia Vilor-Tejedor
Ph.D. Co-Supervisor

Institute for Risk Assessment Sciences, Department of Veterinary Medicine, Utrecht University, Utrecht, Netherlands, BarcelonaBeta Brain Research Center, Pasqual Maragall Foundation, Barcelona, Spain
Email: nvilor@barcelonabeta.org

Professor Po-Chih Kuo
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Department of Computer Science, Ph.D. Program in Biomedical Artificial Intelligence, National Tsing Hua University, Hsinchu, Taiwan
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